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In demand function,

$Q_d = 100 - 5P = 100 - 5(10) = 50$

Hence at

$P = 10, Q_d = 50, Qs = 50.$

Quantity demanded is equal to supply at 50 units when price is ₹10

### Example: 12.4

The market demand curve is given by D = 50 – 5P. Find the maximum price beyond which nobody will buy the commodity.

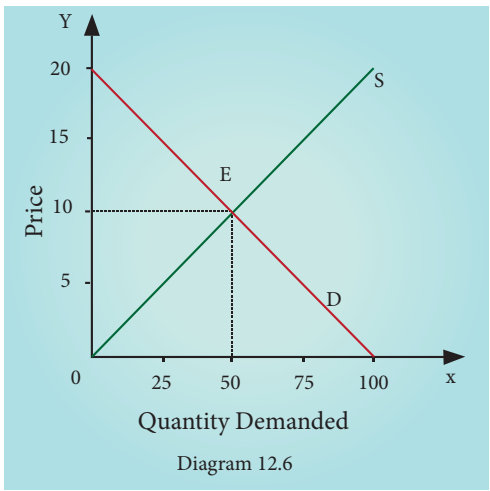


Diagram 12.6

### Solution:

Given

$Q_d = 50 - 5P$

$5P = 50 - Q_d$

$5P = 50$ when $Q_d$ is zero.

$$P = \frac{50}{5}$$

$P = 10$ When $P = 10$, Demand is 0

Hence $P = 10$, which is the maximum price beyond which nobody will demand the commodity.

### Example: 12.5

The demand for milk is given by

| Price (Y) | 1 | 2 | 3 |
|---|---|---|---|
| Demand (X) | 100 | 50 | 0 |

Find the linear demand function and its slope.

### Solution:

Equation of demand function joining two data points (100, 1) and (50, 2) are $(x_1, y_1)$ and $(x_2, y_2)$ respectively.

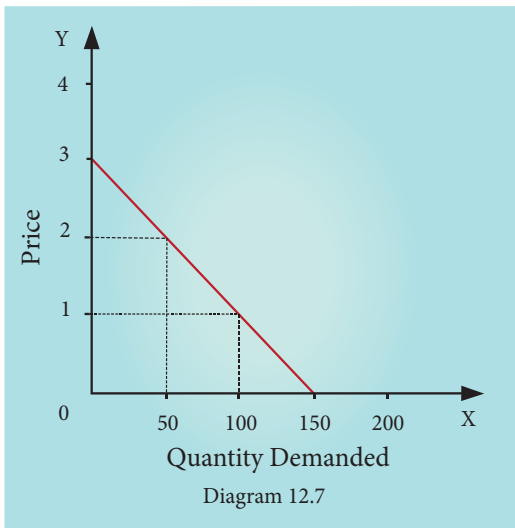


Diagram 12.7

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$$\frac{y - 1}{2 - 1} = \frac{x - 100}{50 - 100}$$

$$\frac{y - 1}{1} = \frac{x - 100}{-50}$$

$-50(y - 1) = 1(x - 100)$

$-50y + 50 = x - 100$

$-50y + 50 + 100 = x$

$-50y + 150 = x$

$x + 150 = 50y$

Hence the demand function is

$Qd = 150 - 50P$ and Slope $m = -50$



**Think and Do for Water Management in your area**

Try to find the demand function for water in your street and the daily total demand for water in litre for all purposes.